

AE 07 - Brexit - Telling Different Stories with Data

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Introduction

In this application exercise, you'll explore how the same data can tell different stories depending on how you visualize it. You'll work with survey data about Brexit opinions across different regions of Great Britain.

The key lesson: The choices you make in creating visualizations (scales, proportions vs. counts, faceting vs. grouping) dramatically affect the story your visualization tells.

In this application exercise, you will:

- Work with factors using `fct_relevel()` and `fct_recode()`
- Create faceted visualizations
- Use free scales in faceting
- Calculate and visualize proportions
- Create grouped bar plots with `position = "dodge"`
- Format axis labels with the `scales` package
- Interpret how different visualizations tell different stories

Estimated time: 30-45 minutes

Getting Started

Fork and Clone Workflow

Quick reminder: Work in YOUR forked `application-exercises` repository.

1. Navigate to the `ae-07-brexit` folder in YOUR fork
 2. Open `ae-07-brexit.Rmd`
 3. Verify the `data` folder exists with `brexit.csv`
 4. Update the YAML with your name
 5. Knit to verify it works
-

Packages

```
library(tidyverse)
library(scales) # For formatting axis labels (we'll use this in Exercise 2)
```

What these packages do:

- **tidyverse:** Data wrangling and visualization
 - **scales:** Formatting numbers, percentages, and dates on plots
-

The Survey

In September 2019, YouGov surveyed 1,639 GB adults with the following question:

In hindsight, do you think Britain was right/wrong to vote to leave EU?

- Right to leave
- Wrong to leave
- Don't know

The data from the survey is in `data/brexit.csv`.

Load the Data

```
brexit <- read_csv("data/brexit.csv")
```

Explore the data:

```
glimpse(brexit)
```

```
## Rows: 1,639
## Columns: 2
## $ opinion <chr> "Right", "Right", "Right", "Right", "Right", "Right", "Right", ~
## $ region <chr> "london", "london", "london", "london", "london", "london", "l~
# See the first few rows
head(brexit)
```

```
## # A tibble: 6 x 2
##   opinion region
```

```
##   <chr>   <chr>
## 1 Right   london
## 2 Right   london
## 3 Right   london
## 4 Right   london
## 5 Right   london
## 6 Right   london

# Count responses by region
brexit %>%
  count(region)

## # A tibble: 5 x 2
##   region          n
##   <chr>         <int>
## 1 london         197
## 2 midlands_wales 354
## 3 north          400
## 4 rest_of_south  547
## 5 scot          141

# Count responses by opinion
brexit %>%
  count(opinion)
```

```
## # A tibble: 3 x 2
##   opinion          n
##   <chr>         <int>
## 1 Don't know    188
## 2 Right        664
## 3 Wrong        787
```

Understanding the data:

- Each row represents one survey respondent
- region: Which region of GB they're from
- opinion: Their response (Wrong, Right, Don't know)

Original Visualization

In the course video, we made the following visualisation. First, we need to clean up the region names and put them in a logical order.

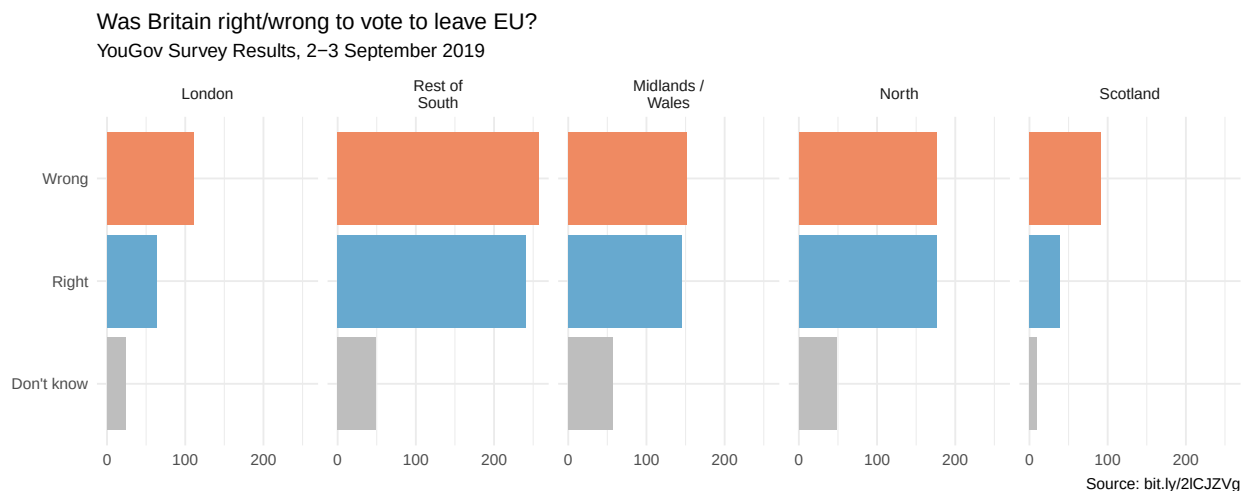
```
brexit <- brexit %>%
  # Reorder regions from south to north
  mutate(
    region = fct_relevel(region, "london", "rest_of_south",
                          "midlands_wales", "north", "scot"),
    # Rename regions with proper capitalization and spacing
    region = fct_recode(region,
                        London = "london",
                        `Rest of South` = "rest_of_south",
                        `Midlands / Wales` = "midlands_wales",
                        North = "north",
                        Scotland = "scot")
  )
```

What this code does:

- `fct_relevel()`: Changes the order of factor levels
- `fct_recode()`: Renames factor levels for better display
- Backticks `Rest of South`: Used for names with spaces

Now create the original plot:

```
ggplot(brexit, aes(y = opinion, fill = opinion)) +  
  geom_bar() +  
  facet_wrap(~region, nrow = 1, labeller = label_wrap_gen(width = 12)) +  
  guides(fill = "none") +  
  labs(  
    title = "Was Britain right/wrong to vote to leave EU?",  
    subtitle = "YouGov Survey Results, 2-3 September 2019",  
    caption = "Source: bit.ly/2lCJZVg",  
    x = NULL,  
    y = NULL  
  ) +  
  scale_fill_manual(values = c(  
    "Wrong" = "#ef8a62",  
    "Right" = "#67a9cf",  
    "Don't know" = "gray"  
  )) +  
  theme_minimal()
```



What this plot shows:

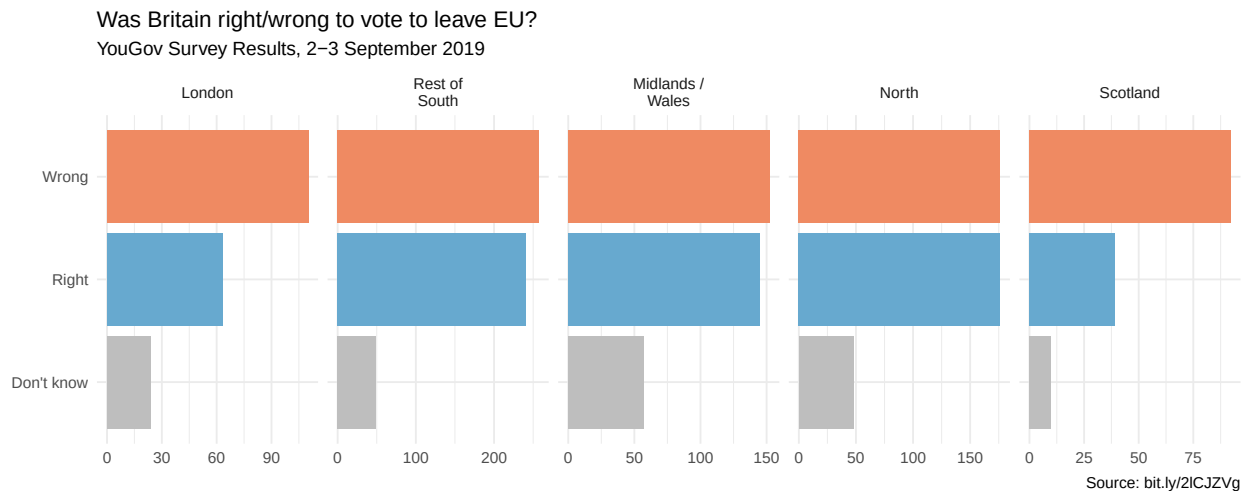
- Counts (not proportions) of each opinion
- Separated by region
- All facets use the same x-axis scale

Exercise 1: Free Scales

Add `scales = "free_x"` as an argument to the `facet_wrap()` function. How does the visualisation change? How is the story this visualisation telling different than the story the original plot tells?

```
# Remember to change eval=FALSE to eval=TRUE!

ggplot(brexit, aes(y = opinion, fill = opinion)) +
  geom_bar() +
  facet_wrap(~region,
    nrow = 1,
    labeller = label_wrap_gen(width = 12),
    scales = "free_x" # Add "free_x" here
  ) +
  guides(fill = "none") +
  labs(
    title = "Was Britain right/wrong to vote to leave EU?",
    subtitle = "YouGov Survey Results, 2-3 September 2019",
    caption = "Source: bit.ly/2lCJZVg",
    x = NULL,
    y = NULL
  ) +
  scale_fill_manual(values = c(
    "Wrong" = "#ef8a62",
    "Right" = "#67a9cf",
    "Don't know" = "gray"
  )) +
  theme_minimal()
```



What `scales = "free_x"` does:

- Each facet gets its own x-axis scale
- The x-axis range is determined independently for each region
- Makes it easier to see proportions within each region

Task 1.1: How does the visualization change with free scales?

Your answer:

Each region gets its own x-axis range, so the bars fill more of each panel. This makes differences between opinions within a region easier to see. The bars still show counts

Task 1.2: How is the story this visualization tells different from the original plot?

Your answer: The free scales plot makes it easier to compare opinions within each region. The original

plot makes it easier to compare response counts across regions because every panel uses the same scale.

Original plot emphasizes: Comparing the actual number of responses across regions using the same scale.

Free scales plot emphasizes: Comparing the relative sizes of the opinion groups within each region.

Task 1.3: Which visualization would be misleading if you wanted to compare the absolute number of responses across regions? Why?

Your answer:

The free scales plot could be misleading when comparing absolute counts across regions. Bars can look equally long even when they represent different numbers because each panel uses a different scale.

Exercise 2: Comparing Proportions Across Facets

First, calculate the proportion of wrong, right, and don't know answers in each category and then plot these proportions (rather than the counts). Then improve axis labeling.

How is the story this visualisation telling different than the story the original plot tells?

Task 2.1: Calculate proportions by region and opinion.

Remember to change eval=FALSE to eval=TRUE!

```
brexit_prop <- brexit %>%  
  count(region, opinion) %>%  
  group_by(region) %>%  
  mutate(prop = n / sum(n))
```

brexit_prop

```
## # A tibble: 15 x 4  
## # Groups:   region [5]  
##   region      opinion      n  prop  
##   <fct>      <chr>    <int> <dbl>  
## 1 London    Don't know     24 0.122  
## 2 London    Right         63 0.320  
## 3 London    Wrong        110 0.558  
## 4 Rest of South Don't know     49 0.0896  
## 5 Rest of South Right        241 0.441  
## 6 Rest of South Wrong        257 0.470  
## 7 Midlands / Wales Don't know     57 0.161  
## 8 Midlands / Wales Right        145 0.410  
## 9 Midlands / Wales Wrong        152 0.429  
## 10 North    Don't know     48 0.12  
## 11 North    Right        176 0.44  
## 12 North    Wrong        176 0.44  
## 13 Scotland Don't know     10 0.0709  
## 14 Scotland Right         39 0.277  
## 15 Scotland Wrong         92 0.652
```

Hints:

- Use `count(region, opinion)`
- Group by `region`
- Calculate proportion: `n / sum(n)`

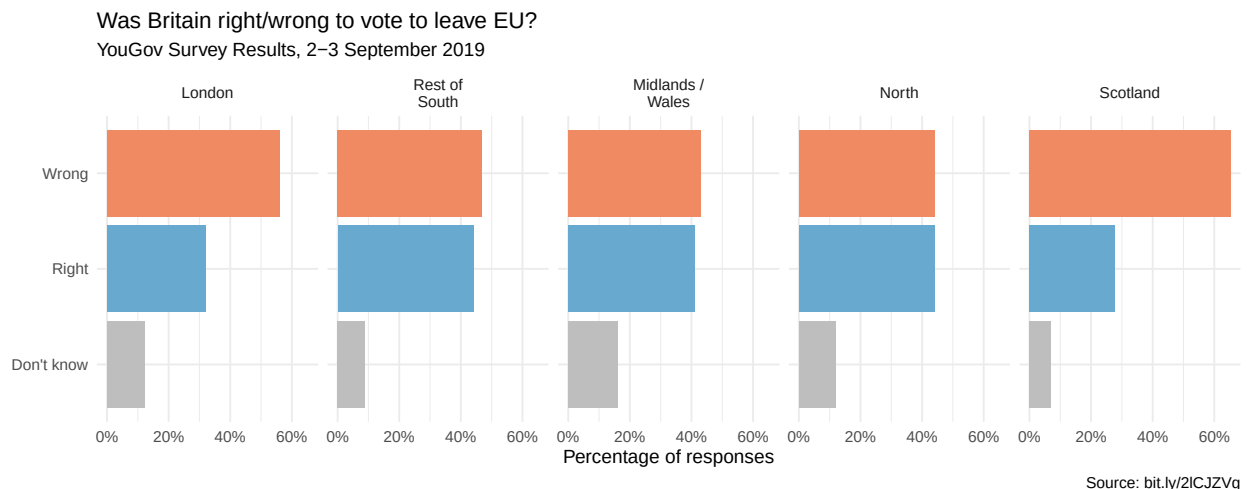
What this creates:

- A new dataset with proportions instead of counts
- Each region's proportions sum to 1 (100%)

Task 2.2: Plot the proportions with improved axis labeling.

Remember to change eval=FALSE to eval=TRUE!

```
ggplot(brexit_prop, aes(y = opinion, x = prop, fill = opinion)) +  
  geom_col() + # Use geom_col() for already-calculated values  
  facet_wrap(~region, nrow = 1, labeller = label_wrap_gen(width = 12)) +  
  guides(fill = "none") +  
  labs(  
    title = "Was Britain right/wrong to vote to leave EU?",  
    subtitle = "YouGov Survey Results, 2-3 September 2019",  
    caption = "Source: bit.ly/2lCJZVg",  
    x = "Percentage of responses", # Label for proportions  
    y = NULL  
  ) +  
  scale_fill_manual(values = c(  
    "Wrong" = "#ef8a62",  
    "Right" = "#67a9cf",  
    "Don't know" = "gray"  
  )) +  
  scale_x_continuous(labels = scales::percent_format()) + # Format x-axis as percentages  
  theme_minimal()
```



Hints:

- x-axis should be prop
- x-axis label: something like "Proportion" or "Percentage"
- For percentage labels: `labels = percent_format()`
- Use `geom_col()` not `geom_bar()` because we already calculated the values

Task 2.3: How is the story this plot tells different from the original?

Your answer: The proportion plot compares the percentage choosing each opinion within each region. This makes regions easier to compare even when they have different numbers of respondents.

Original plot (counts) shows: The number of people who selected each opinion in each region.

Proportion plot shows: The percentage of respondents who selected each opinion within each region.

Which regions have the most “Wrong” responses as a proportion? Scotland, followed by London.

Exercise 3: Comparing Proportions Across Bars

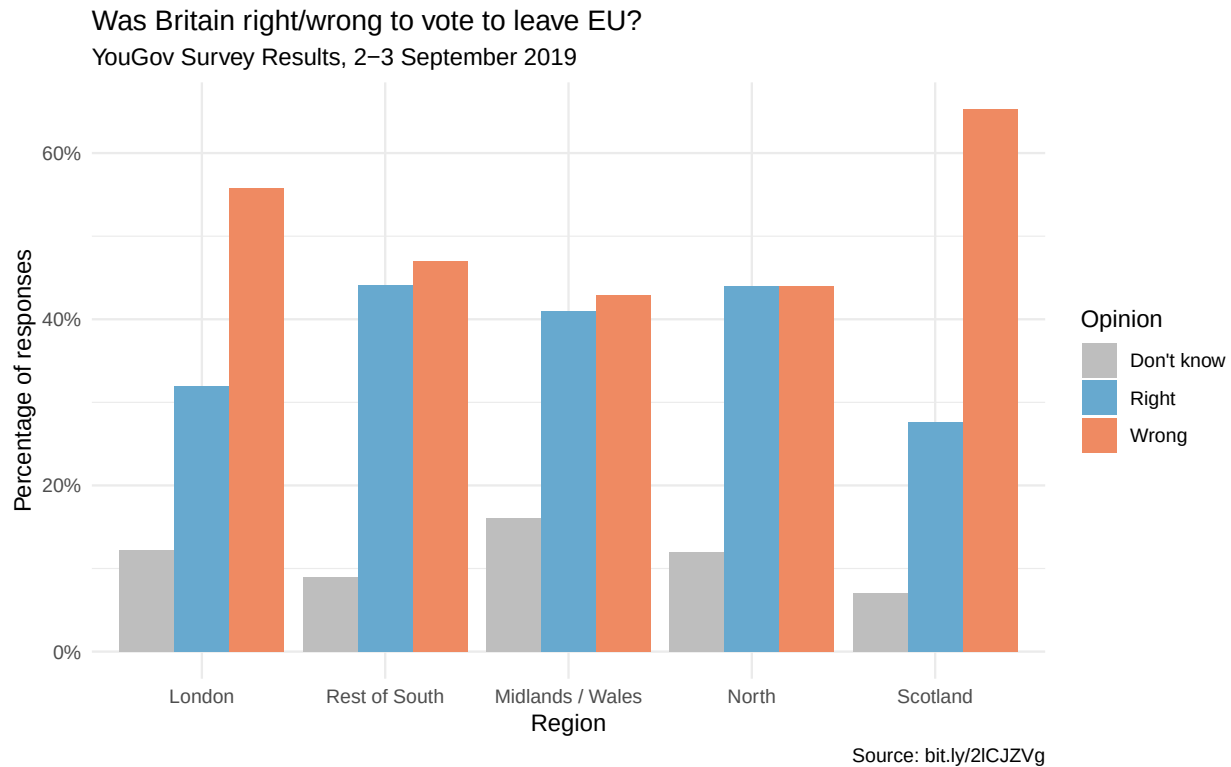
Recreate the same visualisation from the previous exercise, this time dodging the bars for opinion proportions for each region, rather than faceting by region. Then improve the legend.

How is the story this visualisation telling different than the story the previous plot tells?

Task 3.1: Create a dodged bar plot.

```
# Remember to change eval=FALSE to eval=TRUE!

ggplot(brexit_prop, aes(x = region, y = prop, fill = opinion)) +
  geom_col(position = "dodge") + # "dodge" makes bars side-by-side
  labs(
    title = "Was Britain right/wrong to vote to leave EU?",
    subtitle = "YouGov Survey Results, 2-3 September 2019",
    caption = "Source: bit.ly/2lCJZVg",
    x = "Region",
    y = "Percentage of responses",
    fill = "Opinion"
  ) +
  scale_fill_manual(values = c(
    "Wrong" = "#ef8a62",
    "Right" = "#67a9cf",
    "Don't know" = "gray"
  )) +
  scale_y_continuous(labels = scales::percent_format()) +
  theme_minimal()
```



Hints:

- x-axis: `region`
- y-axis: `prop`
- fill: `opinion`
- position: `"dodge"` (makes bars side-by-side)
- x label: `"Region"`
- y label: `"Proportion"`
- fill label: `"Opinion"` or `"Response"`
- y-axis labels: `percent_format()`

Task 3.2: How is this visualization different from the faceted proportion plot?

Your answer: The dodged plot puts all regions on one graph with opinion bars side by side. The faceted plot separates each region into its own panel. Both show the same proportions, but the layout changes how we compare them.

Faceted plot makes it easier to: Focus on the opinion breakdown within each region.

Dodged plot makes it easier to: Compare the percentage choosing the same opinion across regions.

Task 3.3: Which region shows the strongest opinion one way or the other (least “Don’t know”)?

Your answer: Scotland has the lowest percentage of “Don’t know” responses, so it has the largest percentage choosing either “Right” or “Wrong.”

Exercise 4: Reflection on Data Storytelling

Task 4.1: You’re writing a report arguing that Scotland is most opposed to Brexit. Which visualization would you use? Why?

Your answer:

I would use the dodged proportion plot because it shows that Scotland has the highest percentage of “Wrong” responses in this survey. Using percentages makes the comparison fairer because the regions have different numbers of respondents.

Task 4.2: You’re writing a report comparing absolute numbers of respondents across regions. Which visualization would be most appropriate? Why?

Your answer:

I would use the original count plot because all regions share the same scale. This makes it easier to compare the actual numbers of responses across regions.

Task 4.3: What’s one thing you learned about how visualization choices affect the story you tell?

Your answer:

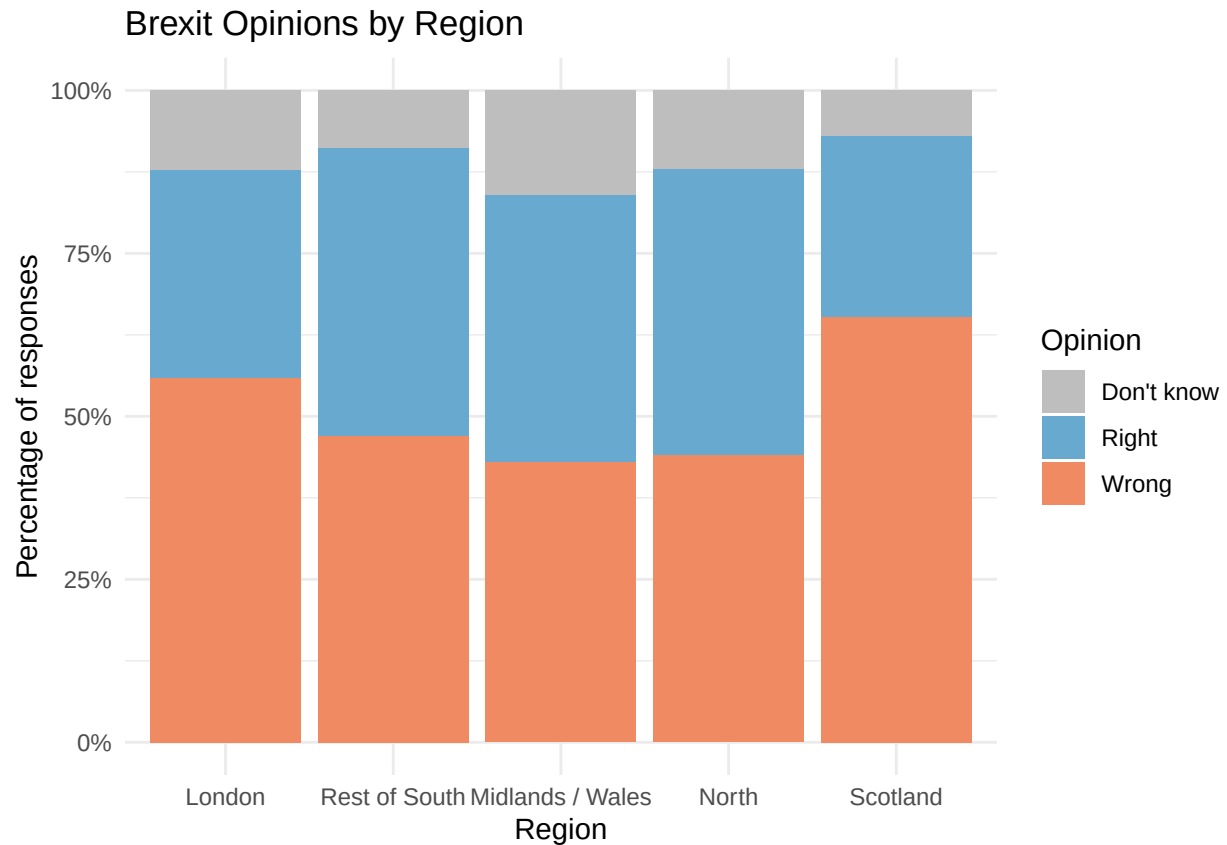
I learned that changing the scale or using percentages can change what stands out in a graph. I should choose a visualization that matches the question and label it clearly so readers understand what it shows.

Exercise 5: Challenge (Optional)

Task 5.1: Create a stacked proportion bar chart where each bar represents a region and is divided by opinion, with each bar going from 0 to 100%.

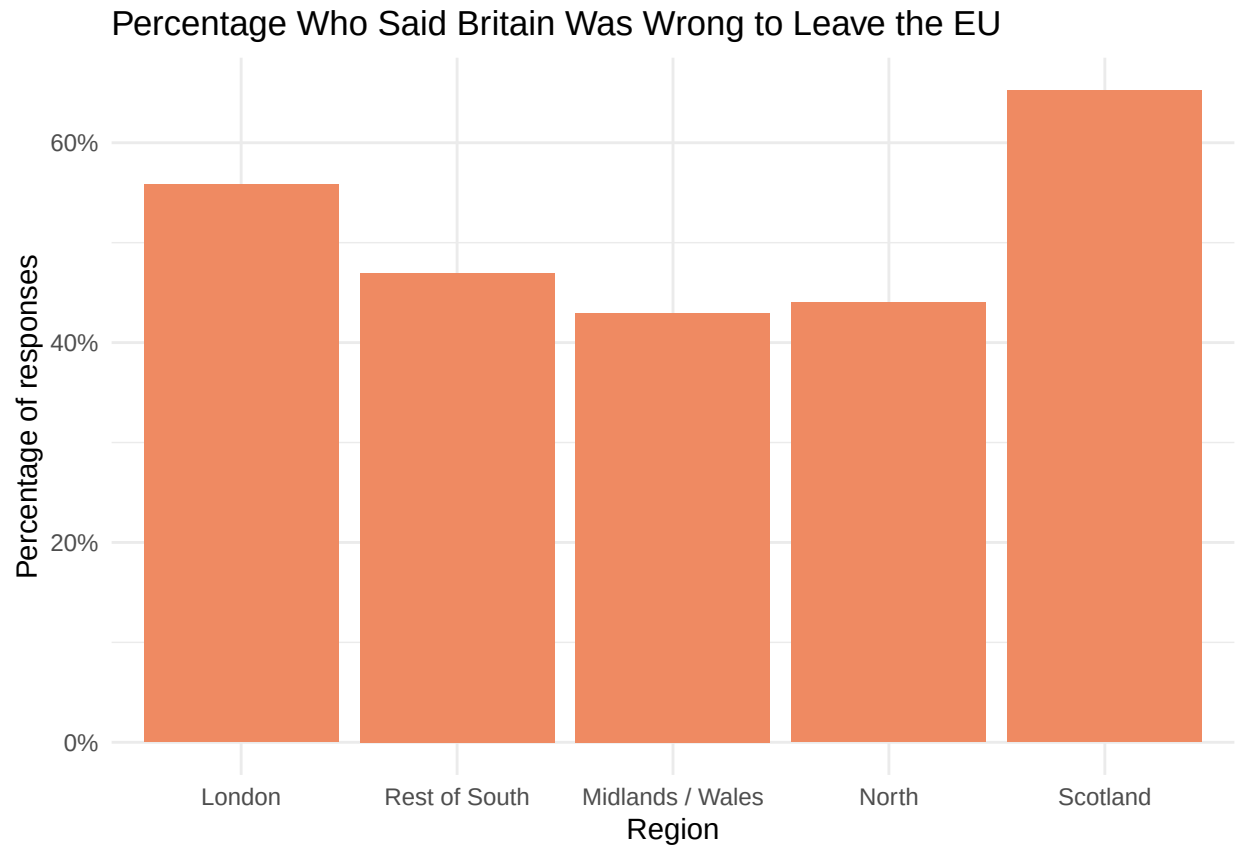
Hint: Use `position = "fill"` in `geom_col()`.

```
ggplot(brexit_prop, aes(x = region, y = prop, fill = opinion)) +  
  geom_col(position = "fill") +  
  labs(  
    title = "Brexit Opinions by Region",  
    x = "Region",  
    y = "Percentage of responses",  
    fill = "Opinion"  
  ) +  
  scale_fill_manual(values = c(  
    "Wrong" = "#ef8a62",  
    "Right" = "#67a9cf",  
    "Don't know" = "gray"  
  )) +  
  scale_y_continuous(labels = scales::percent_format()) +  
  theme_minimal()
```



Task 5.2: Create a visualization showing only the proportion who said “Wrong” across regions.

```
brexit_prop %>%  
  filter(opinion == "Wrong") %>%  
  ggplot(aes(x = region, y = prop)) +  
  geom_col(fill = "#ef8a62") +  
  labs(  
    title = "Percentage Who Said Britain Was Wrong to Leave the EU",  
    x = "Region",  
    y = "Percentage of responses"  
  ) +  
  scale_y_continuous(labels = scales::percent_format()) +  
  theme_minimal()
```



Knit and Submit

Before you finish:

1. Make sure all code chunks run without errors
2. Change `eval=FALSE` to `eval=TRUE` for all exercises
3. Fill in all answer spaces
4. Knit to HTML first to see all visualizations
5. Review all plots - do they tell the story you described?

Git workflow:

1. Click the **Git** pane
 2. Check boxes next to all changed files
 3. Click **Commit**
 4. Write message: "Completed AE 07 - Brexit"
 5. Click **Commit** then **Push**
-

Generate PDF for Canvas Submission

1. Click **Knit** dropdown arrow
2. Select **Knit to PDF**
3. Find `ae-07-brexite.pdf` in Files pane
4. Export and save to your computer

5. Submit to Canvas

Note: The plots might look slightly different in PDF vs HTML due to formatting - this is normal.

Troubleshooting

Common issues:

- **Error: “object ‘bexit_prop’ not found”:** Make sure you ran Exercise 2 first to create this object
- **Percentage formatting not working:** Make sure you loaded the `scales` package
- **Bars not dodging:** Check that `position = "dodge"` is inside `geom_col()`
- **Wrong proportions:** Make sure you grouped by `region` before calculating proportions

Factor issues: - **Regions in wrong order:** The `fct_relevel()` code sets the order - **Region names look weird:** Check the `fct_recode()` code

Visualization issues: - **Can't see all bars:** Adjust `fig.width` and `fig.height` in chunk options - **Legend overlapping plot:** Try `theme(legend.position = "bottom")` - **Facet labels wrapping oddly:** Adjust the width in `label_wrap_gen()`

Scales package: - **percent_format() not working:** Make sure you loaded `library(scales)` - **Percentages showing decimals:** Use `percent_format(accuracy = 1)` for whole numbers

If you're stuck: - Review the walkthrough video - Check ggplot2 documentation: `?facet_wrap`, `?geom_col` - Try each step individually to see where errors occur - Ask on the discussion board

Key Concepts Review

Important visualization concepts from this exercise:

Fixed vs. Free Scales: - Fixed scales: Same axis range across all facets (easier to compare absolute values) - Free scales: Each facet has its own axis range (easier to see patterns within groups)

Counts vs. Proportions: - Counts: Show absolute numbers (good for understanding sample sizes) - Proportions: Show percentages (good for comparing across groups of different sizes)

Faceting vs. Grouping: - Faceting: Separate panels for each category (easier to see patterns within categories) - Grouping (dodged bars): All categories in one panel (easier to compare across categories)

Data storytelling: - The same data can tell different stories depending on how you visualize it - Always consider what story your visualization is telling - Be aware of how your choices might mislead (intentionally or not)

Quick Reference

Factor manipulation:

```
# Reorder levels
fct_relevel(variable, "level1", "level2", ...)

# Rename levels
fct_recode(variable,
            "New Name 1" = "old_name_1",
            "New Name 2" = "old_name_2")
```

Calculate proportions:

```
data %>%  
  count(group_var, count_var) %>%  
  group_by(group_var) %>%  
  mutate(prop = n / sum(n))
```

Faceting:

```
facet_wrap(~variable, scales = "free_x") # Free x-axis  
facet_wrap(~variable, scales = "free_y") # Free y-axis  
facet_wrap(~variable, scales = "free")   # Both axes free
```

Bar positioning:

```
geom_col(position = "dodge") # Side-by-side  
geom_col(position = "stack") # Stacked (default)  
geom_col(position = "fill")  # Stacked to 100%
```

Format as percentages:

```
scale_x_continuous(labels = percent_format())  
scale_y_continuous(labels = percent_format(accuracy = 1)) # Whole numbers
```